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A Theoretical Analysis

A.1 Proof of Theorem 5

Throughout the proof, we will write S to denote a dataset where all of the labels were generated synthetically (*i.e.*, by an LLM), D to denote a dataset where all of the labels were provided by `labeler $_{\beta}$` , and $\mathcal{D}$ to denote a dataset containing a mixture of these kinds of data. Also, only datasets denoted by $\mathcal{D}$ will contain elements whose weights can differ from 0 and 1. All other datasets will be ordinary sets.

We adopt a few simplifying assumptions and conventions. Assume that the given prompt set P is non-empty. Assume that the initial LLM g_0 returns an incorrect response to every prompt. Assume that any LLM returns at most one correct response to any prompt. Removing the latter pair of assumptions would only speed up the convergence of Algorithm 2 to an optimal LLM, but would also further complicate its analysis. Finally, we adopt the convention that $\infty \cdot 0 = 0$. This convention is needed when Algorithm 2 constructs $\mathcal{D}_t$ via the weighted union operation, since it can happen that $\lambda_t = \infty$, but this only occurs when D_t is empty.

Let $P_t^+ = \{x \in P : (x, y) \in S_t^+\}$ be the correct prompts selected by `filter`. By definition P_t^+ and P_t^- form a partition of P . Furthermore, D_t pairs each prompt in P_t^- with the label it was assigned by `labeler $_{\beta}$` , and S_t^+ pairs each prompt in P_t^+ with the (synthetic) label it was assigned by the previous iteration’s LLM, g_{t-1} . Observe that $P_t^-(x) = \sum_y D_t(x, y)$ and $P_t^+(x) = \sum_y S_t^+(x, y)$. For all $t \geq 1$ and $x \in P$ let

$$\begin{aligned} q_t(x) &= \sum_y D_t(y|x)q(x, y) \\ q_t^+(x) &= \sum_y S_t^+(y|x)q(x, y) \\ \bar{q}_t(x) &= \sum_y \mathcal{D}_t(y|x)q(x, y) \end{aligned}$$

be the average quality of the responses to prompt x in datasets D_t , S_t^+ and $\mathcal{D}_t$, respectively. For convenience we also define $\bar{q}_0(x) = \mathbb{E}_{y \sim g_0(x)}[q(x, y)]$. Note that $\bar{q}_0(x) = 0$ for all $x \in \mathcal{X}$ by assumption.

Lemma 6. *For all $t \geq 1$ and $x \in P$*

$$\bar{q}_t(x) = \frac{\sum_{s=1}^t \lambda_s P_s^-(x) q_s(x) + P_s^+(x) q_s^+(x)}{\sum_{s=1}^t \lambda_s P_s^-(x) + P_s^+(x)}.$$

Proof. We have

$$\begin{aligned} \mathcal{D}_t(y|x) &= \frac{\mathcal{D}_t(x, y)}{\sum_{y'} \mathcal{D}_t(x, y')} \\ &= \frac{\sum_{s=1}^t \lambda_s D_s(x, y) + S_s^+(x, y)}{\sum_{y'} \sum_{s=1}^t \lambda_s D_s(x, y') + S_s^+(x, y')} \\ &= \frac{\sum_{s=1}^t \lambda_s D_s(x, y) + S_s^+(x, y)}{\sum_{s=1}^t \lambda_s P_s^-(x) + P_s^+(x)} \end{aligned}$$